A.U. TANEESHA IYENSHI

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OBJECTIVE

Research-oriented graduate, focusing on **Artificial Intelligence, Machine Learning, Biomedical Engineering**and **Cybersecurity**. Passionate about advancing knowledge through experimentation, analysis, and academic research contributions.

EDUCATION

- **B. Sc. (Hons.) in Engineering (Specialized in Computer Engineering)** | **Result Sheets** | **Letter** Mar 2021 - Jan 2026
  Department of Electrical and Information Engineering, University of Ruhuna
  - Current GPA: 3.76/4.00 (First Class Honours)
  - The complete degree program was conducted and assessed in English medium
- **Diploma in Information Technology** | **Certificate** 2020 - 2021
  Siksil Institute of Business and Technology (SIBT)
  - Best Performer of the Batch
- **G.C.E. Advanced Level Examination** | **Result Sheet** 2017 - 2019
  Sirimavo Bandaranaike Vidyalaya - Colombo 07
  - Combined Mathematics, Physics, Chemistry (AAB)

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PATENTS AND PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=SUBMITTED, T=THESIS

[C.1] Aberuwan RMMP, Abesundara WHS, Iyenshi AUT, et al. (2026). **AI Integrated Hand Function Recovery System with Mobile Gaming & Smart Glove for Stroke Rehabilitation**. In *19th Association of Sri Lankan Neurologists Annual Congress (ASNAC 2026)*, pp. 11–12. Association of Sri Lankan Neurologists, Sri Lanka, 26 Feb–3 Mar 2026. DOI: 10.4038/sljon.v13i5.218.

[P.1] Athukoralage Uduli Taneesha Iyenshi, Rajapaksha Pathirannehelage Monilka Ranjanya Rajapaksha, et al. (2026). **A System for Hand Function Recovery Integrated with a Smart Glove, Artificial Intelligence and Mobile Gaming for Stroke Rehabilitation**. Sri Lanka Patent Office, Patent Pending.

[S.1] A.U.T. Iyenshi, R.M.P.R. Rajapaksha (2026). **Comparative Analysis of Intrinsic Glass-Box Models and Post-Hoc XAI Frameworks for Explainable Loan Approval Prediction**. Submitted to *SLAAI 2026*.

[S.2] A.U.T. Iyenshi, K.A. Iroshan, K.L.K. Sudheera (2026). **A Robust and Adaptive Blind Image Watermarking Framework Using DWT-SVD, QIM, and AES Encryption**. Submitted to *ICAC 2026*.

[S.3] I.K.Y.K. Gunawardhana, H.M.L.B. Herath, K.G.M. Levmanthi, G.G.A. Sandaruwan, A.U.T. Iyenshi, et al. (2026). **AI-Driven Cyber Threat Intelligence: Automated Threat Extraction and Analysis from Open and Dark Web Sources**. Submitted to *BuildSEC 2026*.

PROJECTS

- **Neuro-Symbolic Evidence Grounding for CTI Relation Extraction** *Jul 2026 - Present*
  Tools: Python, PyTorch, Hugging Face (Qwen2.5), spaCy, LLaMA 3, Ollama, STIX 2.1, MITRE ATT&CK [🌐]
  - Engineered a neuro-symbolic relation extraction framework combining deterministic syntax parsing, hierarchical sub-span containment guards, and LLM refinement to construct CTI knowledge graphs.
  - Formulated programmatic pre-inference evidence gates that eliminated distant parametric hallucinations (100% suppression) and cut LLM compute by $> 70\%$ while sustaining a 0.9082 USES F1 score.
- **AI-Driven Cyber Threat Intelligence Pipeline (Co-Supervisor)** *Mar 2026 - Aug 2026*
  Tools: Python, PyTorch, Hugging Face, LLaMA 3, Ollama, Neo4j, ChromaDB, PostgreSQL, FastAPI, Tor [🌐]
  - Engineered an automated CTI pipeline extracting threat data from open and dark web sources using SecureBERT-NER, multi-store persistence (Neo4j, ChromaDB, PostgreSQL), and external validation via NVD and AbuseIPDB.
  - Contribution: Co-supervised 4 researchers on system architecture and formulated programmatic grounding constraints to eliminate LLM hallucinations during relation extraction.
- **Blockchain-Based Vehicle Maintenance Tracking System** *Dec 2025 - Feb 2026*
  Tools: Ethereum (Solidity), Node.js, Express, Web3.js, React, Next.js, Ethers.js, IPFS (Pinata), OpenZeppelin [🌐]
  - A blockchain-based system for secure vehicle maintenance tracking with a multi-party approval workflow to ensure transparency and prevent fraud.

◦ Contribution: Developed smart contracts for approval workflows, integrated IPFS for document storage, and built backend APIs and frontend interfaces to support system operations.

- **Genome Matcher – Bioinformatics DNA Analysis Tool** *Oct 2025 - Dec 2025*
  Tools: Haskell (GHC 9.10+), Stack, Functional Programming, Concurrent Processing, File I/O [🌐]
  ◦ A functional programming-based bioinformatics tool that compares DNA sequences to detect mutations and assess genetic risk for medical research applications.
  ◦ Contribution: Designed and implemented pure functional algorithms for DNA comparison, mutation detection, and risk scoring, along with concurrent processing support for analyzing multiple patient genomes efficiently.
- **Financial News Analyzer & Summarizer with MLOps** *Oct 2025 - Dec 2025*
  Tools: Python, Flask, Transformers, Docker, MLflow [🌐]
  ◦ A full-stack application that performs sentiment analysis and text summarization on financial news using optimized retrained transformer models.
  ◦ Contribution: Integrated transformer-based models, deployed them as scalable APIs, containerized services using Docker, and implemented an MLOps pipeline with MLflow for versioning, CI, and automated deployment.
- **AI Integrated Hand Function Recovery System with Mobile Gaming & Smart Glove for Stroke Rehabilitation** *Feb 2025 - Dec 2025*
  Tools: Google MediaPipe, Dart, Flutter
  ◦ A clinical platform for stroke rehabilitation that uses hand biomechanics and mobile gaming to assess and improve patient recovery.
  ◦ Contribution: Developed and optimized scoring algorithms based on hand biomechanics, with a unique, normalized pipeline for hand landmark data integration with mobile gaming. Additionally, assisted in project coordination and documentation.
- **ChromaGuard Pro – Advanced Invisible Image Watermarking System** *Apr 2025 - Oct 2025*
  Tools: Python, Flask, OpenCV, NumPy, PyWavelets (DWT), Cryptography (AES), Pillow, HTML, Tailwind CSS, JavaScript [🌐]
  ◦ A web-based application for embedding and extracting invisible, encrypted watermarks in images using a robust DWT-SVD algorithm.
  ◦ Contribution: Implemented the watermarking algorithm with AES encryption, developed a Flask-based full-stack application, and integrated attack simulation features to test watermark robustness.
- **Heritage Compass – RAG-Based AI Research Assistant** *Jul 2025 - Aug 2025*
  Tools: Python, Flask, LangChain, Qdrant, LLMs, RAG Architecture [🌐]
  ◦ An AI-powered web application that delivers accurate, context-aware answers on UNESCO World Heritage Sites using a Retrieval-Augmented Generation (RAG) framework with vector search and LLMs.
  ◦ Contribution: Developed a reliable data pipeline using Wikidata SPARQL, implemented metadata-based retrieval with embeddings, and built a responsive full-stack interface for multilingual query handling.
- **Vojo Tourism - Software Project** *Oct 2023 - May 2024*
  Tools: Node.js,REST API, MongoDB, Postman, Jira, Agile
  ◦ A unified tourism platform for Sri Lanka that enables trip planning, booking, and seamless connection with local service providers.
  ◦ Contribution: Managed project workflows using Jira and backlogs while developing and testing backend APIs to support core platform functionality.

SKILLS

- **AI & Machine Learning:** PyTorch, TensorFlow, Scikit-learn, Transformers, Deep Learning, Computer Vision, Explainable AI
- **LLMs & NLP:** Hugging Face, LLaMA, Qwen, RAG, LangChain, Ollama, Embeddings, Vector Databases
- **Cybersecurity & CTI:** STIX 2.1, MITRE ATT&CK, Cyber Threat Intelligence, Threat Extraction
- **Programming:** Python, JavaScript, Haskell, Solidity
- **Data & MLOps:** MLflow, Qdrant, Neo4j, PostgreSQL, Docker, Apache Airflow, Apache NiFi
- **Backend & Development:** FastAPI, Flask, REST APIs, Git, Linux

ACADEMIC EXPERIENCE

- **Teaching Assistant** | **Letter** Jan 2026 - Present
  Department of Electrical and Information Engineering, University of Ruhuna
  ◦ Conducting laboratory sessions for undergraduates
    * EE2201 - Fundamentals of Electronics
    * EE4304 / EC4304 - Digital Logic Design
    * EE3203 / EC3203 - Electrical and Electronic Measurements
  ◦ Conducting tutorial classes and lectures

    - EE3202 / EC3202 - Data Structures and Algorithms
    - EE7214 - Information Security
  - Project Co-supervision
    - Co-supervised the final year undergraduate project "AI-Driven Cyber Threat Intelligence: Automated Threat Extraction and Analysis from Open and Dark Web Sources"
  - Supporting assessment, lab evaluations, and student queries

- **Student Instructor** | **Letter** Jan 2025 - Dec 2025
  Department of Electrical and Information Engineering, University of Ruhuna
  - Conducting laboratory sessions for undergraduates
    - EE4304 / EC4304 - Digital Logic Design

Industrial Experience

- **Data Engineer - Contract** | **Service Letter** Aug 2025 - Mar 2026
  Aethria Solutions
  - Designed and implemented a complete automated ETL pipeline from data ingestion to storage.
  - Applied data cleaning, validation, and integrated anomaly detection for identifying irregular patterns.
  - Implemented logging and conducted testing to ensure system reliability and accuracy.
- **Data Science Engineer** Nov 2024 - May 2025
  Freelance
  - Built an end-to-end machine learning pipeline for solar power generation forecasting, including preprocessing, feature engineering, and model training to improve prediction accuracy.
  - Designed and maintained data pipelines and automated reporting workflows, enabling real-time dashboards and supporting data-driven decision-making.
- **Intern - Analytical Platforms** | **Service Letter** Jul 2024 - Dec 2024
  Midas Safety Sri Lanka (Pvt) Ltd
  - Built an end-to-end ML pipeline with automated preprocessing (feature engineering, imputation, scaling), model training, and SHAP-based explainability, with deployment-ready inference.
  - Developed a pipeline to convert unstructured PDFs/images into structured JSON using OCR, image preprocessing, and LLM-based semantic parsing and entity extraction.

Volunteer Experience

- **Secretary** *2025 – 2026*
  IEEE Industrial Application Society, University of Ruhuna
  - Led operations, coordinated events, and improved member engagement through strategic planning and communication.
- **Junior Treasurer** *2024 – 2025*
  IEEE Industrial Application Society, University of Ruhuna
  - Managed budgeting, financial records, and ensured transparent resource allocation for society activities.
- **Publicity Lead – SparkLink 1.0** *Jul 2023 – Sep 2023*
  IEEE Student Branch, University of Ruhuna
  - Executed promotional campaigns to boost event visibility and participant engagement.
- **Membership and Recruitment Lead** *2023 – 2024*
  IEEE Computer Society, University of Ruhuna
  - Drove membership growth through recruitment initiatives and community engagement activities.
- **Event Moderator** *2023 – 2025*
  Career Guidance Unit, IEEE Student Branch, University of Ruhuna
  - Moderated technical and career events, ensuring smooth sessions and effective audience engagement.
- **Event Moderator – INSPIRE Webinar Series** *2023*
  Department of Interdisciplinary Studies, Faculty of Engineering, University of Ruhuna
  - Hosted and coordinated webinar sessions, facilitating discussions and enhancing participant experience.

Extracurricular Activities

- **Chess** *2011 – 2018*
  Sirimavo Bandaranaike Vidyalaya, Colombo 07 | *Samudradevi Balika Vidyalaya, Nugegoda*
  - Represented school Chess team in Inter-school competitions and played in Chess tournaments for the sports meet.
- **Communication & Media Unit** *2013 – 2016*
  Samudradevi Balika Vidyalaya, Nugegoda
  - Active member (2013–2016); involved in announcements and media coordination.
  - Served as President (2016), leading communication activities and team coordination.

References

Dr. Kushan Sudheera
Senior Lecturer, Dept. of Electrical & Info. Eng.,
University of Ruhuna, Sri Lanka.
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Relationship: Project Supervisor

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Relationship: Project Co-supervisor